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Lab 2 : Docker Container Lifecycle and Management Using Nginx

Step 1: Check Docker Installation

docker --version

Output:

```
C:\>docker --version
Docker version 29.4.0, build 9d7ad9f
```

Step 2: Search for Nginx

docker search nginx

Output:

```
C:\>docker search nginx
NAME                DESCRIPTION                STARS     OFFICIAL
nginx               Official build of Nginx.    21337     [OK]
nginx/nginx-ingress NGINX and NGINX Plus Ingress Controllers fo... 121
nginx/nginx-prometheus-exporter NGINX Prometheus Exporter for NGINX and NGIN... 52
nginx/nginx-ingress-operator NGINX Ingress Operator for NGINX and NGINX P... 4
nginx/nginxaas-loadbalancer-kubernetes          1
bitnamicharts/nginx Bitnami Helm chart for NGINX Open Source        4
ubuntu/nginx       Nginx, a high-performance reverse proxy & we... 141
kasmweb/nginx      An Nginx image based off nginx:alpine and in... 9
rancher/nginx      An Nginx container, brought to you by LinuxS... 236
linuxserver/nginx  T-Pot Nginx                                0
dtagdevsec/nginx   0
paketobuildpacks/nginx 3
vmware/nginx       A customized NGINX image containing a consu... 1
gluufederation/nginx Secure by Design, Built for Speed, Hardened ... 0
cleanstart/nginx   Nginx server used for Antrea e2e testing       0
antrea/nginx       ActiveState's customizable, low-to-no vulner... 0
activestate/nginx  0
intel/nginx        Nginx service image for Docksal                1
docksal/nginx      Our customized nginx image                    0
geokrety/nginx     This image is for internal use                 2
circleci/nginx     Nginx customized to run Ilios along with the... 0
ilios/nginx        https://github.com/corpusops/docker-images/   1
corpusops/nginx    Nginx Docker image for PHP development. SSL-... 0
wayofdev/nginx     Nginx SSL / HSTS / HTTP2                      3
dockette/nginx
```

Step 3: Download Nginx

docker pull nginx

Output:

```
C:\>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
45ef33eb91c8: Pull complete
6a24a6ebcf30: Pull complete
3b9f3509a826: Pull complete
062e450697fa: Pull complete
39694cde734e: Pull complete
8d2092eabbc5: Pull complete
0b19be717994: Pull complete
cd64a2895680: Download complete
29456554d205: Download complete
Digest: sha256:b5a9a3cfc86b81dd6f1458d53f2ad2b559cc255653d2efd6cacee5636bf2066f
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

Step 4: View Images

Output:

Step C

Gordon BETA

Containers

Images

Volumes

Kubernetes

Builds

Docker Hub

Docker Scout

Models

MCP Toolkit BETA

Extensions

Containers / gracious_dijkstra

gracious_dijkstra

5dbbd14f2449 nginx:latest

STATUS

Running (15 minutes ago)

Logs

Inspect

Bind mounts

Exec

Files

Stats

/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration

/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/

/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh

10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf

10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf

/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh

/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh

/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh

/docker-entrypoint.sh: Configuration complete; ready for start up

2026/07/14 11:55:05 [notice] 1#1: using the "epoll" event method

2026/07/14 11:55:05 [notice] 1#1: nginx/1.31.2

2026/07/14 11:55:05 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)

2026/07/14 11:55:05 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2

2026/07/14 11:55:05 [notice] 1#1: getrlimit(RLIMIT_NFILE): 1048576:1048576

2026/07/14 11:55:05 [notice] 1#1: start worker processes

2026/07/14 11:55:05 [notice] 1#1: start worker process 29

2026/07/14 11:55:05 [notice] 1#1: start worker process 30

2026/07/14 11:55:05 [notice] 1#1: start worker process 31

2026/07/14 11:55:05 [notice] 1#1: start worker process 32

2026/07/14 11:55:05 [notice] 1#1: start worker process 33

2026/07/14 11:55:05 [notice] 1#1: start worker process 34

2026/07/14 11:55:05 [notice] 1#1: start worker process 35

2026/07/14 11:55:05 [notice] 1#1: start worker process 36

2026/07/14 11:55:05 [notice] 1#1: start worker process 37

2026/07/14 11:55:05 [notice] 1#1: start worker process 38

2026/07/14 11:55:05 [notice] 1#1: start worker process 39

2026/07/14 11:55:05 [notice] 1#1: start worker process 40

Search

Refresh

Copy

Close

Step D:

```

2026/07/14 11:55:05 [notice] 1#1: start worker process 36
2026/07/14 11:55:05 [notice] 1#1: start worker process 37
2026/07/14 11:55:05 [notice] 1#1: start worker process 38
2026/07/14 11:55:05 [notice] 1#1: start worker process 39
2026/07/14 11:55:05 [notice] 1#1: start worker process 40
2026/07/14 12:11:36 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/07/14 12:11:36 [notice] 29#29: exiting
2026/07/14 12:11:36 [notice] 30#30: exiting
2026/07/14 12:11:36 [notice] 30#30: exit
2026/07/14 12:11:36 [notice] 29#29: exit
2026/07/14 12:11:36 [notice] 32#32: exiting
2026/07/14 12:11:36 [notice] 32#32: exit
2026/07/14 12:11:36 [notice] 31#31: exiting
2026/07/14 12:11:36 [notice] 33#33: exiting
2026/07/14 12:11:36 [notice] 33#33: exit
2026/07/14 12:11:36 [notice] 34#34: exiting
2026/07/14 12:11:36 [notice] 34#34: exit
2026/07/14 12:11:36 [notice] 35#35: exiting
2026/07/14 12:11:36 [notice] 35#35: exit
2026/07/14 12:11:36 [notice] 36#36: exiting
2026/07/14 12:11:36 [notice] 36#36: exit
2026/07/14 12:11:36 [notice] 38#38: exiting
2026/07/14 12:11:36 [notice] 38#38: exit
2026/07/14 12:11:36 [notice] 39#39: exiting
2026/07/14 12:11:36 [notice] 39#39: exit
2026/07/14 12:11:36 [notice] 37#37: exiting
2026/07/14 12:11:36 [notice] 37#37: exit
2026/07/14 12:11:36 [notice] 31#31: exit
2026/07/14 12:11:36 [notice] 40#40: exiting
2026/07/14 12:11:36 [notice] 40#40: exit
2026/07/14 12:11:36 [notice] 1#1: signal 14 (SIGALRM) received
2026/07/14 12:11:36 [notice] 1#1: signal 17 (SIGCHLD) received from 29
2026/07/14 12:11:36 [notice] 1#1: worker process 29 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: worker process 30 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: worker process 31 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: worker process 32 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: worker process 35 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: worker process 37 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: worker process 38 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: worker process 36 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: worker process 39 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: worker process 40 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 12:11:36 [notice] 1#1: signal 17 (SIGCHLD) received from 37
2026/07/14 12:11:36 [notice] 1#1: signal 17 (SIGCHLD) received from 33
2026/07/14 12:11:36 [notice] 1#1: worker process 33 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: signal 29 (SIGIO) received
2026/07/14 12:11:36 [notice] 1#1: signal 17 (SIGCHLD) received from 34
2026/07/14 12:11:36 [notice] 1#1: worker process 34 exited with code 0
2026/07/14 12:11:36 [notice] 1#1: exit

```

Step 6: Detached Mode

docker run -d nginx

Output:

```
C:\>docker run -d nginx
0ae133f23912ddda0f8b4350c35c9c3c5f508903748218ff328516344f938bd4
```

Step 7: Name the Container

docker run -d --name mynginx nginx

Output:

```
C:\> docker run -d --name mynginx nginx
16414a8f59392953ecc89c935fae225d60ecf90af3bbf8ebec0c48e3736081fe
```

Step 8: Publish a Port

docker run -d --name webserver -p 8080:80 nginx

Output:

```
C:\>docker run -d --name webserver -p 8080:80 nginx
f5203514b2d5ce23ebc489437eca8bfb7eeb759cc64aacc24ef31a2b1d1f299d
```

Step 9: List Running Containers

docker ps

Output:

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
f5203514b2d5	nginx	"/docker-entrypoint..."	2 minutes ago	Up 2 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	webserver
16414a8f5939	nginx	"/docker-entrypoint..."	3 minutes ago	Up 3 minutes	80/tcp	mynginx
0ae133f23912	nginx	"/docker-entrypoint..."	5 minutes ago	Up 5 minutes	80/tcp	intelligent_bell

Step 10: Stop Container

docker stop mynginx

Output:

```
C:\>docker stop mynginx
mynginx
```

Step 11: Start Container Again

docker start mynginx

Output:

```
C:\>docker start mynginx
mynginx

C:\>|
```

Step 12: Restart

docker restart mynginx

Output:

```
C:\>docker restart mynginx
mynginx

C:\>|
```

Step 13: View Logs

docker logs mynginx

Output:

```
C:\>docker logs mynginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 12:15:15 [notice] 1#1: using the "epoll" event method
2026/07/14 12:15:15 [notice] 1#1: nginx/1.31.2
2026/07/14 12:15:15 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 12:15:15 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/07/14 12:15:15 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 12:15:15 [notice] 1#1: start worker processes
2026/07/14 12:15:15 [notice] 1#1: start worker process 29
2026/07/14 12:15:15 [notice] 1#1: start worker process 30
2026/07/14 12:15:15 [notice] 1#1: start worker process 31
2026/07/14 12:15:15 [notice] 1#1: start worker process 32
2026/07/14 12:15:15 [notice] 1#1: start worker process 33
2026/07/14 12:15:15 [notice] 1#1: start worker process 34
2026/07/14 12:15:15 [notice] 1#1: start worker process 35
2026/07/14 12:15:15 [notice] 1#1: start worker process 36
2026/07/14 12:15:15 [notice] 1#1: start worker process 37
2026/07/14 12:15:15 [notice] 1#1: start worker process 38
2026/07/14 12:15:15 [notice] 1#1: start worker process 39
2026/07/14 12:15:15 [notice] 1#1: start worker process 40
2026/07/14 12:19:45 [notice] 1#1: signal 3 (SIGQUIT) received, shutting down
2026/07/14 12:19:45 [notice] 29#29: gracefully shutting down
2026/07/14 12:19:45 [notice] 32#32: gracefully shutting down
2026/07/14 12:19:45 [notice] 29#29: exiting
2026/07/14 12:19:45 [notice] 32#32: exiting
2026/07/14 12:19:45 [notice] 34#34: gracefully shutting down
2026/07/14 12:19:45 [notice] 29#29: exit
2026/07/14 12:19:45 [notice] 31#31: gracefully shutting down
2026/07/14 12:19:45 [notice] 33#33: gracefully shutting down
2026/07/14 12:19:45 [notice] 31#31: exiting
2026/07/14 12:19:45 [notice] 33#33: exiting
2026/07/14 12:19:45 [notice] 32#32: exit
2026/07/14 12:19:45 [notice] 33#33: exit
2026/07/14 12:19:45 [notice] 31#31: exit
2026/07/14 12:19:45 [notice] 35#35: gracefully shutting down
2026/07/14 12:19:45 [notice] 35#35: exiting
2026/07/14 12:19:45 [notice] 36#36: gracefully shutting down
2026/07/14 12:19:45 [notice] 39#39: gracefully shutting down
2026/07/14 12:19:45 [notice] 35#35: exit
2026/07/14 12:19:45 [notice] 36#36: exiting
2026/07/14 12:19:45 [notice] 39#39: exiting
2026/07/14 12:19:45 [notice] 36#36: exit
2026/07/14 12:19:45 [notice] 39#39: exit
2026/07/14 12:19:45 [notice] 37#37: gracefully shutting down
2026/07/14 12:19:45 [notice] 37#37: exiting
2026/07/14 12:19:45 [notice] 30#30: gracefully shutting down
2026/07/14 12:19:45 [notice] 30#30: exiting
2026/07/14 12:19:45 [notice] 30#30: exit
2026/07/14 12:19:45 [notice] 40#40: gracefully shutting down
2026/07/14 12:19:45 [notice] 38#38: gracefully shutting down
2026/07/14 12:19:45 [notice] 40#40: exiting
2026/07/14 12:19:45 [notice] 38#38: exiting
2026/07/14 12:19:45 [notice] 40#40: exit
2026/07/14 12:19:45 [notice] 38#38: exit
2026/07/14 12:19:45 [notice] 37#37: exit
```

Step 14: Enter the Container

docker exec -it mynginx bash

Output:

```
C:\>docker exec -it mynginx bash
root@16414a8f5939:/# pwd
/
root@16414a8f5939:/# ls
bin    docker-entrypoint.d  home    media  proc  sbin  tmp
boot   docker-entrypoint.sh lib      mnt    root  srv   usr
dev    etc                  lib64   opt    run   sys   var
root@16414a8f5939:/#
```

Step 15: Remove the Container

docker rm mynginx

Docker deletes the **container only**.

Container

 Deleted

The image is still available.

You can create a new container anytime using:

`docker run nginx`

Output:

```
C:\>docker rm mynginx
mynginx
```

Step 16: Remove the Image

`docker rmi nginx`

Now Docker removes the **downloaded image** from your local system.

Image

 Deleted

If any container still depends on that image, Docker will refuse to remove it until those containers are deleted.

Output:

```
C:\>docker rmi nginx
Error response from daemon: conflict: unable to delete nginx:latest (must be forced) - container 5dbbd14f2449 is using its referenced image b5a9a3cfc86b
C:\>
```